10
Flow Measurement

In this chapter we discuss the various methods and instruments used in the measurement of liquid that flows through a pipeline. The formulas used for calculating the liquid velocities, flow rates, etc., from the pressure readings, their limitations, and the degree of accuracy attainable will be covered for some of the more commonly used instruments. Considerable work is currently being done in this field of flow measurement to improve the accuracy of instruments, particularly when custody transfer of products is involved. For more detailed analysis of the various flow measurement devices, the reader is referred to the publications listed in the References section.

10.1 History

Measurement of flow of liquids has been going on for centuries. The Romans used some form of flow measurement to allocate water from the aqueducts to the houses in their cities. This was necessary to control the quantity of water used by the citizens and prevent waste. Similarly, it is reported that the Chinese used to measure salt water to the brine pots that were used for salt production. In later years, a commodity had to be measured so that it could be properly allocated and the ultimate user charged for it appropriately. Today, gasoline is dispensed from meters at gas stations and the recipient is billed according to the volume of gasoline so measured. Water companies measure water consumed by a household using water meters, while natural gas for residential and industrial consumers is measured by gas meters. In all these instances, the objective is to ensure that the supplier gets paid for the commodity and the user of the commodity is billed for the product at the agreed price. In addition, in industrial processes that involve the use of liquids and gases to perform a specific function, accurate quantities need to be dispensed so that the desired effect of the processes may be realized. In most cases involving consumers, certain regulatory or public agencies (for example, the department of weights and measures) periodically check flow measurement devices to ensure that they are performing accurately and if necessary calibrate them against a very accurate master device.

10.2 Flow Meters

Several types of instruments are available to measure the flow rate of a liquid in a pipeline. Some measure the velocity of flow, while others directly measure the volume flow rate or the mass flow rate.

The following flow meters are used in the pipeline industry:

Venturi meter
Flow nozzle
Orifice meter
Flow tube
Rotameter
Turbine meter
Vortex flow meter
Magnetic flow meter
Ultrasonic flow meter
Positive displacement meter
Mass flow meter
Pitot tube

The first four items listed above are called variable-head meters since the flow rate measured is based on the pressure drop due to a restriction in the meter, which varies with the flow rate. The last item, the Pitot tube, is also called a velocity probe, since it actually measures the velocity of the liquid.

From the measured velocity, we can calculate the flow rate using the conservation of mass equation, discussed in an earlier chapter.

$$\text{Mass flow} = (\text{Flow rate}) \times (\text{density}) = (\text{Area}) \times (\text{Velocity}) \times (\text{density})$$

or

$$M = Q_\rho = AV_\rho$$

Since the liquid density is practically constant, we can state that:

$$Q = AV \tag{10.2}$$

where
A = Cross-sectional area of flow
V = Velocity of flow
ρ = Liquid density

We will discuss the principle of operation and the formulas used for some of the more common meters described above. For a more detailed discussion on flow meters and flow measurement, the reader is referred to one of the fine texts listed in the References section.

10.3 Venturi Meter

The Venturi meter, also known as a Venturi tube, belongs to the category of variable-head flow meters. The principle of a Venturi meter is depicted in Figure 10.1. This type of a Venturi meter is also known as the Herschel type and consists of a smooth gradual contraction from the main pipe size to the throat section, followed by a smooth, gradual enlargement from the throat section to the original pipe diameter.

The included angle from the main pipe to the throat section in the gradual contraction is generally in the range of 21°±2°. Similarly, the gradual expansion from the throat to the main pipe section is limited to a range of 5° to 15° in this design of Venturi meter. This construction results in minimum energy loss, causing the discharge coefficient (discussed later) to approach the value of 1.0. This type of a Venturi meter is generally rough cast with a pipe diameter range of 4.0 in. to 48.0 in. The throat diameter may vary considerably, but the ratio of the throat diameter to the main pipe diameter (d/D), also known as the beta ratio, represented by the symbol ß, should range between 0.30 and 0.75.

The Venturi meter consists of a main piece of pipe which decreases in size to a section called the throat, followed by a gradually increasing size back to the original pipe size. The liquid pressure at the main pipe section

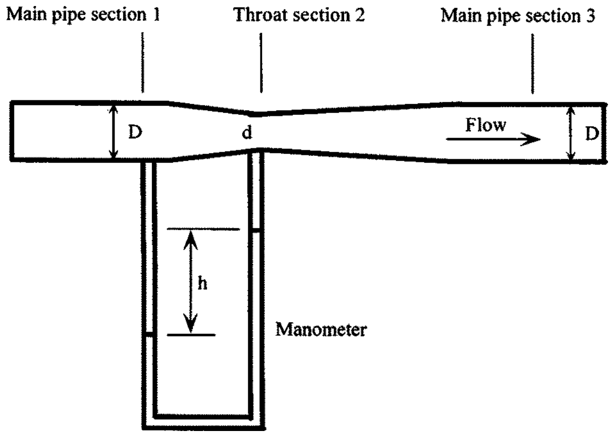

Figure 10.1 Venturi meter.

1 is denoted by P_1 and that at the throat section 2 is represented by P_2. As the liquid flows through the narrow throat section, it accelerates to compensate for the reduction in area, since the volume flow rate is constant (Q=AV from Equation 10.2). As the velocity increases in the throat section, the pressure decreases (Bernoulli's equation). As the flow continues past the throat, the gradual increase in the flow area results in a reduction in flow velocity back to the level at section 1; correspondingly the liquid pressure will increase to some value higher than at the throat section.

We can calculate the flow rate using Bernoulli's equation, introduced in Chapter 2, along with the continuity equation, based on the conservation of mass as shown in Equation (10.1). If we consider the main pipe section 1 and the throat section 2 as reference, and apply Bernoulli's equation for these two sections, we get

$$P_1/\gamma + V_1^2/(2g) + Z_1 = P_2/\gamma + V_2^2/(2g) + Z_2 + h_L \tag{10.3}$$

And from the continuity equation

$$Q = A_1 V_1 = A_2 V_2 \tag{10.4}$$

where γ is the specific weight of liquid and h_L represents the pressure drop due to friction between sections 1 and 2.

Simplifying the above equations yields the following:

$$V_1 = \sqrt{[2g(P_1 - P_2)/\gamma + (Z_1 - Z_2) - h_L]/[(A_1/A_2)^2 - 1]} \tag{10.5}$$

The elevation difference Z_1-Z_2 is negligible even if the Venturi meter is positioned vertically. We will also drop the friction loss term h_L and include it in a coefficient C, called the discharge coefficient, and rewrite Equation (10.5) as follows:

$$V_1 = C\sqrt{[2g(P_1 - P_2)/\gamma]/[(A_1/A_2)^2 - 1]} \tag{10.6}$$

The above equation gives us the velocity of the liquid in the main pipe section 1. Similarly, the velocity in the throat section, V_2, can be calculated using Equation (10.4) as follows:

$$V_2 = C\sqrt{[2g(P_1 - P_2)/\gamma]/[1 - (A_2/A_1)^2]} \tag{10.7}$$

The volume flow rate Q can now be calculated using Equation (10.4) as follows:

$$Q = A_1V_1$$

Therefore,

$$Q = CA_1\sqrt{[2g(P_1 - P_2)/\gamma]/[(A_1/A_2)^2 - 1]} \tag{10.8}$$

Since the beta ratio $\beta = d/D$ and $A_1/A_2 = (D/d)^2$ we can write the above equations in terms of the beta ratio as follows:

$$Q = CA_1\sqrt{[2g(P_1 - P_2)/\gamma]/[(1/\beta)^4 - 1]} \tag{10.9}$$

It can be seen by examining Equations (10.5), (10.6) and (10.7) that the discharge coefficient C actually represents the ratio of the velocity of liquid through the Venturi meter to the ideal velocity when the energy loss (h_L) is zero. C must therefore be a number less than 1.0. The value of C depends on the Reynolds number in the main pipe section 1 and is shown graphically in Figure 10.2. For a Reynolds number greater than 2×10^5 the value of C remains constant at 0.984.

For smaller piping (2 to 10 in.), Venturi meters are machined and hence have a better surface finish than the larger, rough cast meters. These smaller Venturi meters have a C value of 0.995 for Reynolds number greater than 2×10^5.

10.4 Flow Nozzle

A typical flow nozzle is illustrated in Figure 10.3. It consists of the main pipe section 1, followed by a gradual reduction in flow area and a subsequent short cylindrical section, and finally an expansion to the main pipe section 3.

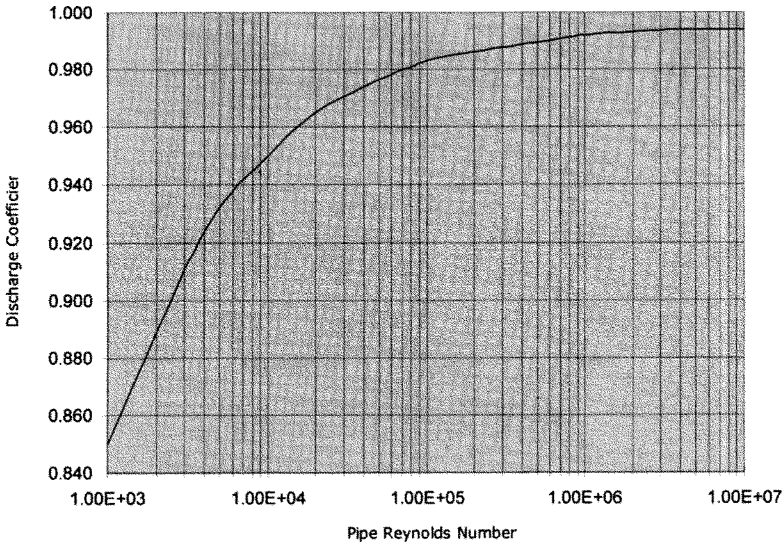

Figure 10.2 Venturi meter discharge coefficient.

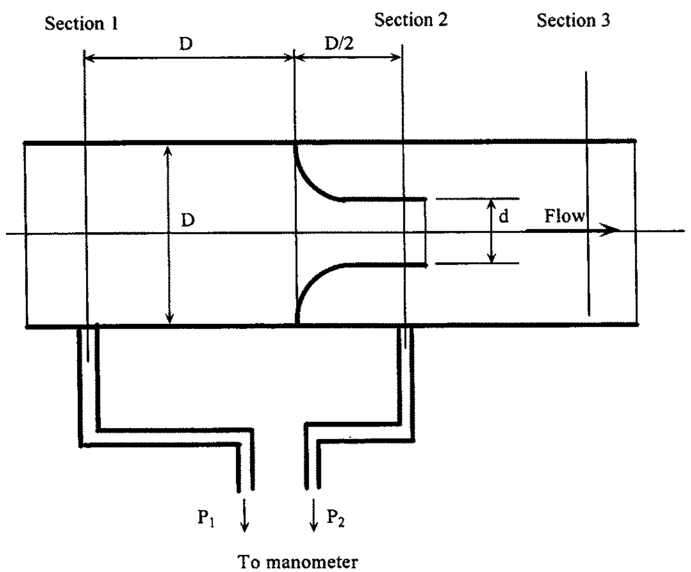

Figure 10.3 Flow nozzle. (From Mott, Robert L., *Applied Fluid Mechanics, 5th ed.,* Copyright 2000. Reprinted by permission of Pearson Education, Inc., Upper Saddle River, NJ.)

The American Society of Mechanical Engineers (ASME) and the International Standards Organization (ISO) have defined the geometries of these flow nozzles and published equations to be used with them. Due to the smooth gradual contraction from the main pipe diameter D to the nozzle diameter d, the energy loss between sections 1 and 2 is very small. We can apply the same Equations (10.6) through (10.8) for the Venturi meter to the flow nozzle also. The discharge coefficient C for the flow nozzle is found to be 0.99 or better for Reynolds numbers above 10^6. At lower Reynolds numbers there is greater energy loss immediately following the nozzle throat, due to sudden expansion, and hence C values are lower.

Depending on the beta ratio and the Reynolds number, the discharge coefficient C can be calculated from the equation

$$C = 0.9975 - 6.53\sqrt{(\beta/R)} \tag{10.10}$$

where

$\beta = d/D$

and R is the Reynolds number based on the pipe diameter D.

Compared with the Venturi meter, the flow nozzle is more compact, since it does not require the length for a gradual decrease in diameter at the throat, or the additional length for the smooth, gradual expansion from the throat to the main pipe size. However, there is more energy loss (and, therefore, pressure head loss) in the flow nozzle, due to the sudden expansion from the nozzle diameter to the main pipe diameter. The latter causes greater turbulence and eddies compared with the gradual expansion in the Venturi meter.

10.5 Orifice Meter

An orifice meter consists of a flat plate that has a sharp-edged hole accurately machined in it and placed concentrically in a pipe, as shown in Figure 10.4. As liquid flows through the pipe, the flow suddenly contracts as it approaches the orifice and then suddenly expands after the orifice back to the full pipe diameter. This forms a vena contracta or a throat immediately past the orifice. This reduction in flow pattern at the vena contracta causes increased velocity and hence lower pressure at the throat, similar to the Venturi meter discussed earlier.

The pressure difference between section 1, with the full flow, and section 2 at the throat can then be used to measure the liquid flow rate, using equations developed earlier for the Venturi meter and the flow nozzle. Due to the sudden contraction at the orifice and the subsequent sudden

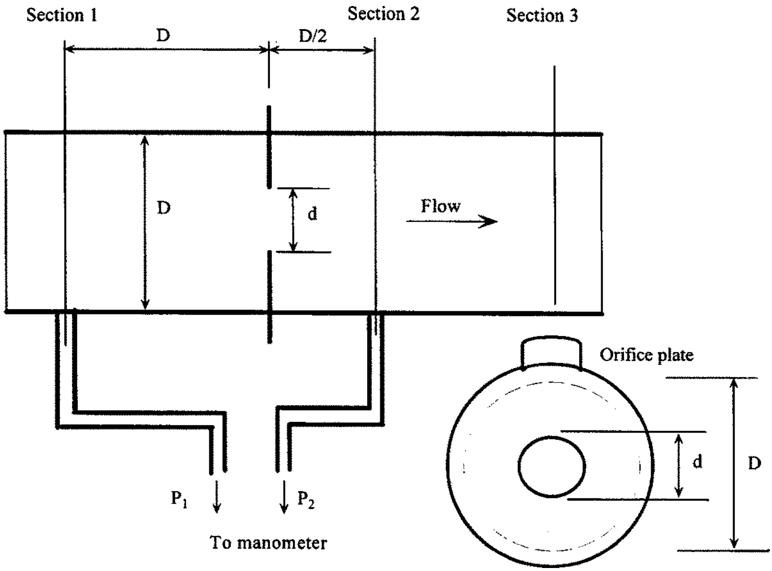

Figure 10.4 Orifice meter. (From Mott, Robert L., *Applied Fluid Mechanics, 5th ed.,* Copyright 2000. Reprinted by permission of Pearson Education, Inc., Upper Saddle River, NJ.)

Table 10.1 Pressure Taps for Orifice Meter

	Inlet pressure tap, P_1	Outlet pressure tap, P_2
1	One pipe diameter upstream from plate	One-half pipe diameter downstream of inlet face of plate
2	One pipe diameter upstream from plate	At vena contracta
3	Flange taps, 1 in. upstream from plate	Flange taps, 1 in. downstream from outlet face of plate

expansion after the orifice, the coefficient of discharge C for the orifice meter is much lower than that of a Venturi meter or a flow nozzle. In addition, depending on the pressure tap locations in section 1 and section 2, the value of C is different for orifice meters.

There are three possible pressure tap locations for an orifice meter, as listed in Table 10.1. Figure 10.5 shows the variation of C with the beta ratio d/D for various values of pipe Reynolds number.

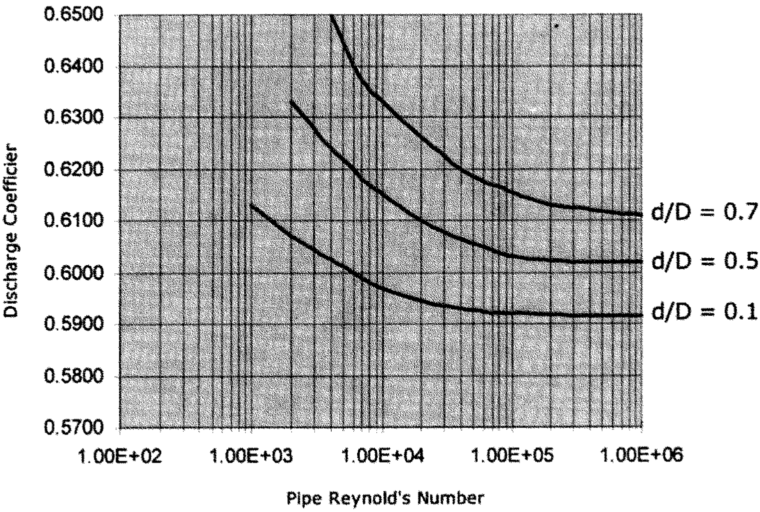

Figure 10.5 Orifice meter discharge coefficient. (From Mott, Robert L., *Applied Fluid Mechanics, 5th ed.,* Copyright 2000. Reprinted by permission of Pearson Education, Inc., Upper Saddle River, NJ.)

Comparing the three types of flow meter discussed above, we can conclude that the orifice meter has the highest energy loss due to the sudden contraction followed by the sudden expansion. On the other hand, the Venturi meter has a lower energy loss compared with the flow nozzle due to the smooth, gradual reduction at the throat followed by the smooth gradual expansion after the throat.

Flow tubes are proprietary, variable-head flow meters that are streamlined in design to cause the least amount of energy loss. They are manufactured by different companies with their own proprietary designs.

10.6 Turbine Meters

Turbine meters are used in a wide variety of applications. The food industry uses turbine meters for measurement of milk, cheese, cream, syrups, vegetable oils, etc. Turbine meters are also used in the oil industry.

Basically the turbine meter is a velocity-measuring instrument. Liquid flows through a free-turning rotor that is mounted co-axially inside the meter. Upstream and downstream of the meter, certain lengths of straight piping are required to ensure smooth velocities through the meter. The liquid striking the rotor causes it to rotate, the velocity of rotation being proportional to the flow rate. From the rotor speed measured and the flow area, we can compute the flow rate through the meter. The turbine meter must be calibrated since flow depends on the friction, turbulence, and the manufacturing tolerance of the rotor parts.

For liquids with viscosity close to that of water the range of flow rates is 10:1. With higher- and lower-viscosity liquids it drops to 3:1. Density effect is similar to viscosity.

A turbine meter requires a straight length of pipe before and after the meter. The straightening vanes located in the straight lengths of pipe helps eliminate swirl in flow. The upstream length of straight pipe has to be 10D in length, where D is the pipe diameter. After the meter, the straight length of pipe is 5D long. Turbine meters may be of a two-section type or three-section type, depending upon the number of pieces the meter assembly is composed of. For liquid flow measurement, the API *Manual of Petroleum Measurement* standard must be followed. In addition, bypass piping must also be installed to isolate the meter for maintenance and repair. To maintain flow measurements on a continuous basis for custody transfer, an entire spare meter unit will be required on the bypass piping, to be used when the main meter is taken out of service for testing and maintenance.

From the turbine meter reading at flowing temperature, the flow rate at some base temperature (such as 60°F) is calculated using the following equation:

$$Q_b = Q_f \times M_f \times F_t \times F_p \tag{10.11}$$

where

Q_b=Flow rate at base conditions, such as 60°F and 14.7 psi

Q_f=Measured flow rate at operating conditions, such as 80°F and 350 psi

M_f=Meter factor for correcting meter reading, based on meter calibration data

F_t=Temperature correction factor for correcting from flowing temperature to the base temperature

F_p=Pressure correction factor for correcting from flowing pressure to the base pressure

10.7 Positive Displacement Meter

The positive displacement (PD) meter is most commonly used in residential applications for measuring water and natural gas consumption. PD meters can measure small flows with good accuracy. These meters are used when consistently high accuracy is required under steady flow through a pipeline. PD meters are accurate within ±1% over a flow range of 20:1. They are suitable for batch operations, for mixing and blending of liquids that are clean without deposits, and for use with noncorrosive liquids.

PD meters basically operate by measuring fixed quantities of the liquid in containers that are alternately filled and emptied. The individual volumes are then totaled and displayed. As the liquid flows through the meter, the force from the flowing stream causes a pressure drop in the meter, which is a function of the internal geometry. The PD meter does not require the straightening vanes as with a turbine meter and hence is a more compact design. However PD meters can be large and heavy compared with an equivalent turbine meter. Also, jamming can occur when the meter is stopped and restarted. This necessitates some form of bypass piping with valves to prevent a pressure rise and damage to meter. The accuracy of a PD meter depends on clearance between the moving and stationary components. Hence precision machined parts are required to maintain accuracy. PD meters are not suitable for slurries or liquids with suspended particles that could jam the components and cause damage to the meter. There are several types of PD meter, including the reciprocating piston type, rotating disk, rotary piston, sliding vane, and rotary vane type.

Many modern petroleum installations employ PD meters for crude oil and refined products. These have to be calibrated periodically, to ensure accurate flow measurement. Most PD meters used in the oil industry are tested at regular intervals, such as daily or weekly, using a master meter called a meter prover system that is installed as a fixed unit along with the PD meter.

Example Problem 10.1

A Venturi meter is used for measuring the flow rate of water at 70° F. The flow enters a 6.625 in. pipe, 0.250 in. wall thickness; the throat diameter is 2.4 in. The Venturi meter is of the Herschel type and is rough cast with a mercury manometer. The manometer reading shows a pressure difference of 12.2 in. of mercury. Calculate the flow velocity in the pipe and the volume flow rate. Use a specific gravity of 13.54 for mercury.

Solution

At 70°F the specific weight and viscosity are:

$$\gamma = 62.3 \text{ lb/ft}^3$$

and

$$v=1.05\times10^{-5}\ ft^2/s$$

Assume first that the Reynolds number is greater than 2×10^5. Therefore, the discharge coefficient C will be 0.984 from Figure 10.2. The beta ratio $\beta=2.4/(6.625-0.5)=0.3918$ and therefore β is between 0.3 and 0.75.

$$A_1/A_2=(1/0.3918)^2=6.5144$$

The manometer reading will give us the pressure difference, by equating the pressures at the two points in the manometer:

$$P_1+\gamma_w(y+h)=P_2+\gamma_w y+\gamma_m h$$

where y is depth of the higher mercury level below the centerline of the Venturi meter and γ_w and γ_m are the specific weights of water and mercury respectively.
Simplifying the above, we get:

$$(P_1-P_2)/\gamma w=(\gamma_m/\gamma_w-1)h$$
$$=(13.54\times62.4/62.3-1)\times12.2/12=12.77\ ft$$

And using Equation (10.6), we get

$$V_1=0.984\sqrt{[(64.4\times12.77)/(6.5144^2-1)]}$$
$$=4.38\ ft/s$$

Now we check the value of the Reynolds number:

$$R=4.38\times(6.125/12)/1.05\times10^{-5}$$
$$=2.13\times10^5$$

Since R is greater than 2×10^5 our assumption for C is correct. The volume flow rate is

$$Q=A_1V_1$$
$$\text{Flow rate}=0.7854(6.125/12)^2\times4.38=0.8962\ ft^3/s$$

or

$$\text{Flow rate}=0.8962\times(1728/231)\times60=402.25\ gal/min$$

Example Problem 10.2

An orifice meter is used to measure the flow rate of kerosene in a 2 in. schedule 40 pipe at 70°F. The orifice is 1.0 in. diameter. Calculate the volume flow rate if the pressure difference across the orifice is 0.54 psi. Specific gravity of kerosene=0.815 and viscosity=2.14×10^{-5} ft^2/s.

Solution

Since the flow rate depends on the discharge coefficient C, which in turn depends on the beta ratio and the Reynolds number, we will have to solve this problem by trial and error.

The 2 in. schedule 40 pipe is 2.375 in. outside diameter and 0.154 in. wall thickness. The beta ratio

$$\beta=1.0/(2.375-2\times0.154)=0.4838$$

First, we assume a value for the discharge coefficient C (say 0.61) and calculate the flow rate from Equation (10.6). We then calculate the Reynolds number and obtain a more correct value of C from Figure 10.5. Repeating the method a couple of times will yield a more accurate value of C and finally the flow rate.

The density of kerosene=0.815×62.3=50.77 lb/ft^3

Ratio of areas

$$A_1/A_2=(1/0.4838)^2=4.2717$$

Using Equation (10.6), we get

$$V_1 = 0.61\sqrt{[(64.4 \times 0.54 \times 144/(50.77)/(4.2717^2 - 1)]} = 1.4588 \text{ ft/s}$$

$$\text{Reynolds number} = 1.4588(2.067/12)/(2.14 \times 10^{-5}) = 11{,}742$$

Using this value of the Reynolds number, we obtain from Figure 10.5 the discharge coefficient C=0.612. Since we earlier assumed C=0.61 we were not too far off.

Recalculating based on the new value of C:

$$V_1 = 0.612\sqrt{[(64.4 \times 0.54 \times 144/(50.77)/(4.2717^2 - 1)]} = 1.46 \text{ ft/s}$$

$$\text{Reynolds number} = 1.46(2.067/12)/(2.14 \times 10^{-5}) = 11{,}751$$

This is quite close to the previous value of the Reynolds number. Therefore we will use the last calculated value for velocity. The volume flow rate is

$$Q=A_1V_1$$

$$\text{Flow rate}=0.7854(2.067/12)^2\times1.46=0.034\ \text{ft}^3/\text{s}$$

or

$$\text{Flow rate}=0.034\times(1728/231)\times60=15.27\ \text{gal/min}$$

10.8 Summary

In this chapter we discussed the more commonly used devices for the measurement of liquid pipeline flow rates. Variable-head flow meters, such as the Venturi meter, flow nozzle and orifice meter, and the equations for calculating the velocities and flow rate from the pressure drop were explained. The importance of the discharge coefficient and how it varies with the Reynolds number and the beta ratio were also discussed. A trial-and-error method for calculating the flow rate through an orifice meter was illustrated using an example. For a more detailed description and analysis of flow meters, the reader should refer to any of the standard texts used in the industry. Some of these are listed in the References section.

10.9 Problems

10.9.1 Water flow through a 4 in. internal diameter pipeline is measured using a Venturi tube. The flow rate expected is 300 gal/min. Specific weight of water is 62.4 lb/ft^3 and viscosity is 1.2×10^{-5} ft2/s.

(a) Calculate the upstream Reynolds number.
(b) If the pressure differential across the meter is 10 psi, determine the dimensions of the meter.
(c) Calculate the throat Reynolds number.

10.9.2 A 12 in. Venturi meter with a 7 in. throat is used for measuring flow of diesel at 60°F. The differential pressure reads 70 in. of water. Calculate the volume and mass flow rates. Specific gravity of diesel is 0.85 and viscosity is 5 cSt.

10.9.3 An orifice meter with a beta ratio of 0.70 is used for measuring crude oil (0.895 specific gravity and 15 cSt viscosity). The pipeline is 10.75 in. diameter, 0.250 in. wall thickness and the line pressure is 250 psig. With pipe taps the differential pressure is 200 in. of water. Calculate the flow rate of crude oil.

10.9.4 Water flows through a Venturi tube of 300 mm main pipe diameter and 150 mm throat diameter at 150 m^3/hr. The differential manometer deflection is 106 cm. The specific gravity of the manometer liquid is 1.3. Calculate the discharge coefficient of the meter.